FlavourBench: Ranking Frontier Language Models with Executable Culinary Ground Truth

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Abstract

Most open-ended language-model benchmarks inherit a judge: a human preference panel, another model, or a brittle exact-match key. We introduce FlavourBench, an automated benchmark in which a versioned culinary system supplies dense, executable ground truth. Each task presents eight ingredients and asks for a three-ingredient portfolio. Before any evaluated model is called, Epicure scores all 56 possible portfolios. The benchmark contains 640 tasks balanced across substitution, pairing, dietary constraints, and regional composition. We evaluate 20 frontier endpoints on 12800 primary model–task cells and 1280 label-permuted repeats. The FlavourBench Score is the equal-family mean of the frozen per-task scores; failed or invalid responses remain in the denominator at zero. We report 50000 family-stratified bootstrap replicates, simultaneous 95% score bands, all 190 paired model contrasts with Holm control, exact-chance tests, and ingredient-set repeatability. Gemini 3.1 Pro has the highest point estimate at 72.9 (simultaneous 95% CI 69.7–76.1); the data resolve 102 of 190 model pairs. The release includes every prompt, response, route, score map, completion state, and content hash, together with an offline verifier that reconstructs the leaderboard, tables, and figures.

1 Introduction

Benchmarks are strongest when the evaluator can be executed. Code can be tested, database states can be inspected, and games can be scored. Culinary reasoning is usually evaluated with a model judge, a handful of human preferences, or factual questions whose answers do not capture the quality of a complete decision. That leaves two problems: the judge is entangled with the systems being tested, and exact match throws away useful differences between plausible answers.

FlavourBench turns a culinary runtime into a benchmark environment. Epicure represents 1,790 ingredients in a 300-dimensional space and exposes deterministic operations for substitution, pairing, dietary feasibility, and regional composition [12, 13]. A task compiler converts these operations into three-of-eight selection problems. It enumerates every candidate portfolio and freezes a continuous 0–100 score map before model execution. An answer is therefore scored by table lookup, without a model judge and without post-hoc interpretation.

This setup separates the reference system from the evaluated systems. Epicure is not a leaderboard entry and is not "100% accurate." It defines the task-specific reward surface. A model receives 100 on a task only if it selects that task's highest-scoring portfolio; other valid portfolios receive partial credit. The benchmark measures agreement with a published executable culinary environment, not universal human taste.

We use that environment to evaluate a common panel of 20 current endpoints. Every model faces the same 640 tasks and 56 frozen answer scores per task. The design has enough shared tasks to estimate continuous score differences directly rather than infer a ranking from sparse pairwise votes. A separate label-permutation panel measures whether a model's ingredient decision survives superficial answer-position changes.

The paper makes three contributions:

1. an executable culinary benchmark with 35840 model-independent portfolio scores and a single interpretable 0–100 metric;
2. a 20-model, 640-task frontier evaluation with simultaneous uncertainty, paired multiplicity-controlled tests, completion-aware eligibility, and repeatability; and
3. a content-addressed release in which every public number traces to frozen tasks, exact routes, raw responses, and a dependency-light offline scorer.

2 Executable culinary ground truth

2.1 A reference environment, not a model judge

The benchmark binds an exact Epicure release, evidence bundle, application build, ingredient count, and embedding dimensionality by hash. Changing any of these inputs creates a new benchmark task-set identity. This is analogous to fixing a simulator or a code-test suite: the reference is explicit and reproducible, but its external validity remains a scientific question rather than an assumption hidden inside a judge prompt.

For each task t, let $\mathcal{A}_t$ contain all 56 three-item subsets of eight candidates. Epicure assigns a raw utility $u_t(a)$ to every valid $a \in \mathcal{A}_t$. Invalid portfolios under a dietary

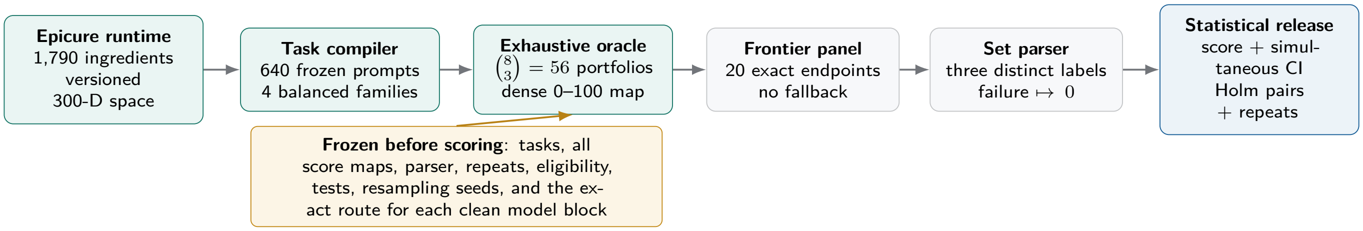

Figure 1: FlavourBench compiles a versioned culinary runtime into dense selection tasks. The evaluated models never judge one another and do not access Epicure. All score maps and the analysis contract are fixed before the primary collection; each route is fixed before its clean scored block, and superseded transport checks remain excluded.

or category constraint receive zero. Valid utilities are normalized within the task:

$$s_t(a) = 100\,\frac{u_t(a) - \min_{b\in\mathcal{A}_t} u_t(b)}{\max_{b\in\mathcal{A}_t} u_t(b) - \min_{b\in\mathcal{A}_t} u_t(b)}. \tag{1}$$

Every task has a unique 100-point optimum. The exact chance baseline is not a generic percentage; it is $|\mathcal{A}_t|^{-1}\sum_a s_t(a)$ for that frozen task.

2.2 Four balanced decision families

The 640 tasks contain 160 items from each family and unique anchor ingredients. Candidate sets are selected across four validation strata per family, then ordered and labeled by a task-derived hash. No evaluated model writes an item, distractor, or score.

The score maps have substantial resolution rather than one correct key plus 55 equivalent mistakes. Table 2 reports the exact random-portfolio baseline, separation of the best and second-best portfolios, and the number of distinct score values. Constraint tasks are intentionally sparse because infeasible portfolios score zero; the other families provide much finer graded rewards.

2.3 Prompt and parser contract

Prompts show the decision context and eight labeled candidates. The final line must contain `FINAL_SELECTION: A,C,F` with three distinct labels. The parser takes the final exact marker line, uppercases the labels, sorts them as an unordered set, and performs one lookup in the frozen score map. Explanatory text does not affect the result. Missing markers, duplicate labels, provider failures, and abnormal finishes score zero.

The model-only condition is deliberate. Epicure supplies the ground truth but is not exposed as a tool during ranking. Thus the score measures whether a model can make the culinary decision, not whether it can copy a runtime return value after being told which operation to call.

3 Evaluation design

3.1 One primary score

For model m and task t, let Y_{mt} be the frozen portfolio score, with $Y_{mt} = 0$ for every failed or invalid response. For family f with task set T_f, the FlavourBench Score is

$$F_m = \frac{1}{4}\sum_{f=1}^{4}\frac{1}{|T_f|}\sum_{t\in T_f} Y_{mt}. \tag{2}$$

It ranges from 0 to 100 and has a direct interpretation: mean quality of the model's selected portfolio under the published Epicure task distribution. Because each family has 160 tasks, Equation (2) also equals the ordinary mean across all 640 tasks.

Completion is reported separately. A model must complete at least 608 of 640 primary requests (95%) to enter the ranked panel. Below that threshold it is marked DNF rather than treated as the least capable model. Within an eligible model, however, every failed or unparseable cell remains a zero in Equation (2); there is no complete-case inflation.

3.2 Frontier endpoint panel

The panel spans OpenAI, Anthropic, Google, xAI, Meta, Kimi, Qwen, Z.ai, DeepSeek, MiniMax, NVIDIA, Mistral, Tencent, and Cohere. It includes GPT-5.6 Sol, Terra, and Luna; Claude Opus and Sonnet 5; Gemini 3.1 Pro and 3.6 Flash; Grok 4.5; Llama 4 Maverick; Kimi K3; Qwen 3.8 Max; GLM 5.2; DeepSeek V4 Pro and Flash; MiniMax M3; Nemotron 3.5 Lightning; Mistral Large 3; Tencent HY 3; Cohere Command A; and Cohere Command R+ (08-2024).

All scored blocks use exact OpenRouter routes. Automatic fallback is disabled. The two Cohere successor blocks are restricted to OpenRouter's Cohere endpoint; they replace, rather than pool with, earlier direct-account calibration responses that hit a monthly request ceiling. The manifest binds the requested model, canonical identity, endpoint tag, provider, supported parameters, price envelope, and execution contract. For endpoints that advertise a reasoning-effort control, hidden reasoning is excluded and the effort is fixed to the lowest stable setting:

Family	Decision	Epicure utility	Hard constraint	Tasks
Substitution	choose a three-item replacement portfolio for an anchor ingredient	0.8 anchor similarity +0.2 portfolio coherence	none	160
Pairing	choose three ingredients to accompany an anchor	0.65 anchor affinity +0.35 portfolio coherence	none	160
Constraints	choose a feasible portfolio under diet and processing limits	0.7 anchor similarity +0.3 coherence	diet and maximum NOVA level	160
Regional composition	choose a category-diverse portfolio for a target cuisine label	0.75 regional direction +0.25 coherence	three distinct ingredient categories	160

Table 1: The four task families share one response and scoring interface but probe different culinary decisions. Each task contains eight candidates and 56 frozen portfolio scores.

Family	Chance	Top gap	Unique scores
Substitution	45.1	6.0	56
Pairing	44.7	6.3	56
Constraints	3.5	35.8	4
Regional composition	36.9	1.7	44

Table 2: Frozen task-map diagnostics for 160 tasks per family, computed before model execution. Chance is the uniformly random portfolio mean; top gap is the median best–second-best margin.

`none` for DeepSeek V4 Flash and GLM 5.2, and `minimal` for the other twelve controllable routes. The remaining six endpoints use their documented provider-fixed behavior. This prevents hidden deliberation from consuming the answer budget while leaving the scored response format identical across routes. The complete route table appears in Appendix A.

3.3 Label-permutation repeatability

Before primary execution, we select 16 tasks per family for a 64-task repeat panel. Candidate labels are cyclically permuted while ingredient identities and all 56 utilities remain unchanged. For each model we compare the original and repeated ingredient sets, not their answer letters. The primary repeat metric is Jaccard agreement; exact set agreement and absolute score change are secondary. Repeats never enter F_m. A model can be declared a definitive top model only if it beats every other eligible model after Holm correction and has mean repeat Jaccard at least 0.80.

3.4 Uncertainty and multiplicity

All inferential choices were fixed before the clean primary run.

- **Scores.** 50000 shared-task bootstrap samples are drawn within each family. We report pointwise percentile intervals and a simultaneous max-t band across all 20 model scores.
- **Pairs.** Every one of the 190 model pairs uses the 640 paired task differences. Two-sided sign-flip tests use 100000 Monte Carlo draws and Holm correction at familywise $\alpha = .05$. We also report paired mean differences, bootstrap intervals, and Cohen's d_z.
- **Chance.** Each model is compared with the exact taskwise mean over all 56 portfolios. The 20 paired sign-flip tests form a separate Holm family.
- **Ranks.** Point ranks are accompanied by bootstrap rank intervals and contiguous statistical groups. Models that are not significantly separated are not presented as a resolved total order.

The preregistered approximation gives greater than 0.99 power for a five-point paired difference at paired task SD 20 after a 190-comparison Bonferroni threshold. Final claims rely on the observed paired tests and bootstrap precision, not on this planning approximation.

4 Results

4.1 A statistically resolved FlavourBench leaderboard

20 of 20 endpoints meet the completion threshold. Gemini 3.1 Pro has the largest score point estimate, 72.9, with simultaneous 95% interval [69.7, 76.1]. Across all 190 paired contrasts, 102 remain significant after Holm correction. Figure 2 shows the score scale, simultaneous intervals, and rank groups; Table 3 gives exact values.

4.2 Which pairwise conclusions survive correction?

The score is common-task and continuous, so each model pair yields 640 matched differences rather than a small number of arena votes. Figure 3 visualizes the multiplicity-controlled results. Grey cells are scientifically useful: they mark pairs for which this task panel does not support a directional claim at familywise $\alpha = .05$. This is different from declaring the two models equal.

4.3 Aggregate scores conceal different culinary profiles

Family-level results in Figure 4 show that similar aggregate scores can arise from different strengths. Constraint tasks penalize infeasible portfolios directly, while pairing and substitution reward graded semantic structure.

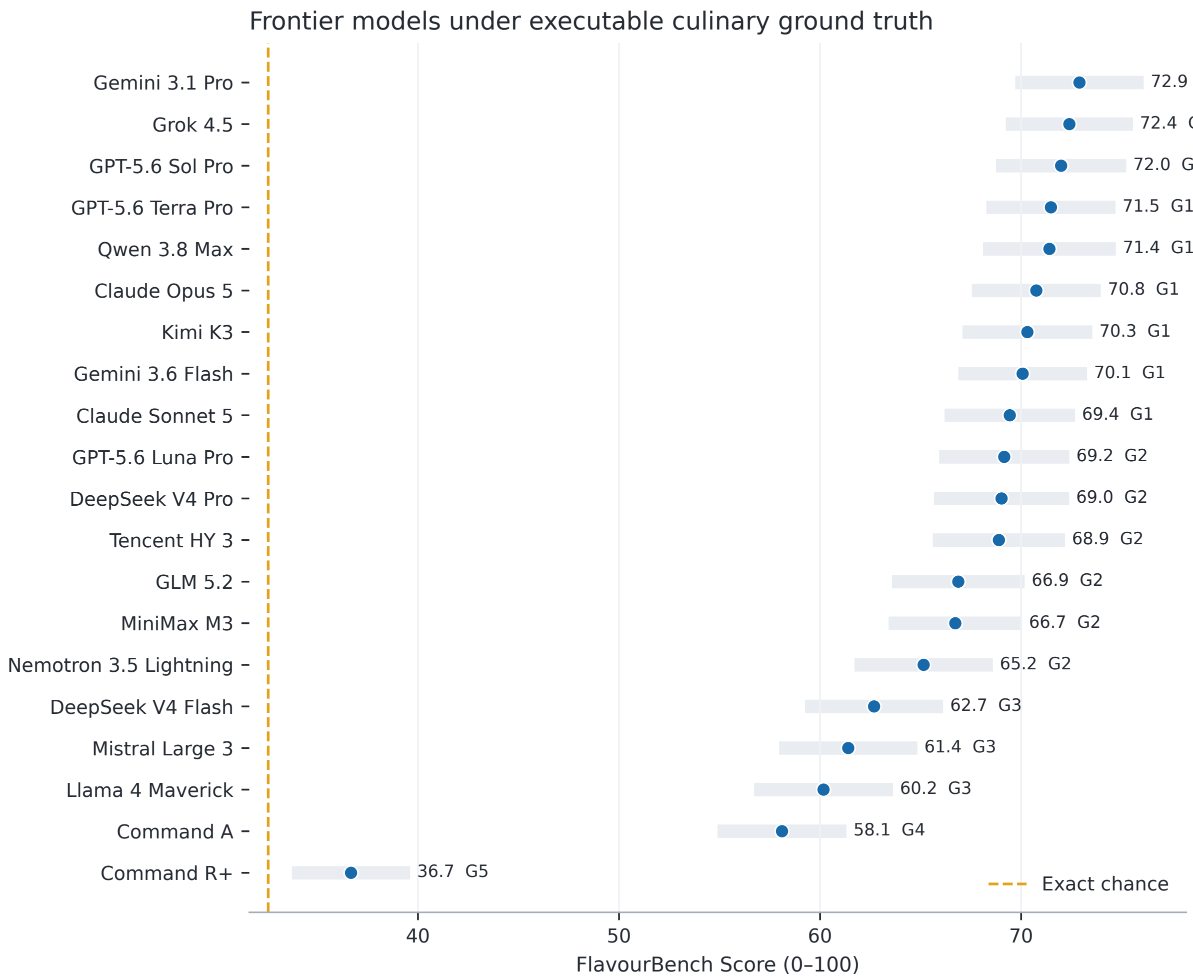

Figure 2: FlavourBench Score with simultaneous 95% max-t intervals. The letter-free group numbers are inferential tiers: a point rank is shown for navigation, while models inside an unresolved group should not be read as a statistically established ordering.

Regional composition adds a diversity constraint and a cuisine-direction utility. The four columns are therefore related but not interchangeable.

4.4 Score and stability answer different questions

The label-permutation panel tests a narrow invariance: does the endpoint choose the same ingredients when answer positions change? Gemini 3.1 Pro has mean ingredient-set Jaccard 0.81. Figure 5 shows that a high culinary score does not mechanically imply high decision stability. This distinction matters for deployment and for training, where a reward can improve while policy variance remains high.

4.5 Real prompts and model decisions

Table 4 reports the lexicographically first frozen task in each family—a selection rule independent of the outcomes. The public case-study JSON contains the complete prompt, all eight choices, the optimal portfolio, both raw model responses, selected ingredient identities, and exact scores. The examples illustrate why continuous scoring is preferable to exact match: two non-optimal portfolios can differ substantially in culinary utility.

5 From benchmark to training signal

The exhaustive task map makes FlavourBench useful beyond a static leaderboard. For a task state x_t and selected portfolio a, Epicure supplies a deterministic reward $r(x_t, a) = s_t(a)$. This creates three immediate training interfaces:

1. supervised learning from the optimal portfolio and the full ordering of 56 candidates;
2. preference or contrastive learning from any pair (a_i, a_j) with known reward difference;
3. offline or online policy optimization using dense

Rank	Model	FB Score	simultaneous 95% CI	group	completed	repeat
1	Gemini 3.1 Pro	72.9	[69.7, 76.1]	1	640/640	0.81
2	Grok 4.5	72.4	[69.2, 75.6]	1	640/640	0.79
3	GPT-5.6 Sol Pro	72.0	[68.7, 75.2]	1	640/640	0.86
4	GPT-5.6 Terra Pro	71.5	[68.3, 74.7]	1	640/640	0.80
5	Qwen 3.8 Max	71.4	[68.1, 74.7]	1	640/640	0.80
6	Claude Opus 5	70.8	[67.5, 74.0]	1	640/640	0.75
7	Kimi K3	70.3	[67.1, 73.5]	1	640/640	0.74
8	Gemini 3.6 Flash	70.1	[66.9, 73.3]	1	640/640	0.80
9	Claude Sonnet 5	69.4	[66.2, 72.7]	1	640/640	0.79
10	GPT-5.6 Luna Pro	69.2	[65.9, 72.4]	2	639/640	0.72
11	DeepSeek V4 Pro	69.0	[65.7, 72.4]	2	634/640	0.71
12	Tencent HY 3	68.9	[65.6, 72.2]	2	640/640	0.52
13	GLM 5.2	66.9	[63.6, 70.2]	2	635/640	0.67
14	MiniMax M3	66.7	[63.4, 70.0]	2	627/640	0.70
15	Nemotron 3.5 Lightning	65.2	[61.7, 68.6]	2	629/640	0.76
16	DeepSeek V4 Flash	62.7	[59.3, 66.1]	3	640/640	0.56
17	Mistral Large 3	61.4	[58.0, 64.8]	3	640/640	0.64
18	Llama 4 Maverick	60.2	[56.7, 63.6]	3	638/640	0.53
19	Command A	58.1	[54.9, 61.3]	4	640/640	0.57
20	Command R+	36.7	[33.7, 39.6]	5	640/640	0.23

Table 3: The complete automated leaderboard. "Repeat" is ingredient-set Jaccard agreement on 64 label-permuted tasks and does not enter the score. Failed and invalid primary responses are included at zero; completion is also shown explicitly.

Family	Prompt	Higher-ranked response	Lower-ranked response
Substitution	Select three alternatives to 'gluten free flour' that best preserve its grain role, regional context, and mutual portfolio coherence.	wheat, buckwheat, wheat germ (50)	cereal, wheat, sponge cake (44)
Pairing	Select the three-ingredient bundle with the strongest learned pairing to 'cannellini bean' and the best internal coherence.	basil, red pepper, tomato (97)	white truffle, barley, wheat (42)
Constraints	For a dish centered on 'muscovado sugar', select three vegetarian complements with NOVA processing level at most 3; among valid portfolios, maximize learned pairing and coherence.	blackberry, bran, graham flour (63)	almond milk, orange liqueur, coffee liqueur (0)
Regional composition	Target cuisine label: Western Atlantic. Select three ingredients that best fit this culinary profile. The selected ingredients must represent three distinct primary roles.	crescent roll, ranch dressing, colby cheese (100)	oyster mushroom, tofu skin, ranch dressing (22)

Table 4: Four outcome-independent examples. Parentheses give the frozen task score. The complete prompts and unabridged responses are released in machine-readable form.

bounded rewards and held-out anchors.

The benchmark release does not claim a reinforcement-learning result. It supplies the environment, reward function, train/dev/test identifiers, and evaluation statistics needed to run one without introducing a model judge. A convincing training experiment should freeze a disjoint ingredient or anchor split, train only on the development reward maps, and report held-out FlavourBench Score, repeatability, and family transfer. Optimizing directly on the public test map would be benchmark memorization, not culinary learning.

6 Reproducibility

The release is organized around content-addressed inputs and append-only outputs. It includes:

- the exact 640-task JSON, with prompts, candidates, all 56 portfolio scores, oracle provenance, task strata, and task-set digest;
- the 20-model manifest, exact provider route per model, fallback policy, supported parameters, endpoint contract, and route digest;
- 12800 primary and 1280 repeat response records, including raw text, completion state, usage, latency, parser output, task score, and response digest;
- the frozen analysis plan, seeds, 50,000 bootstrap and 100,000 sign-flip counts, all 190 pairwise rows, and the final statistical release; and
- a verifier and asset builder that reconstruct the leaderboard CSV, pairwise CSV, LaTeX tables, and vector figures from local files without provider access.

Responses are written once under a model slot and cell identifier. Existing valid cells are skipped on resume; conflicting artifacts fail closed. Provider fallback is disabled, so the route shown in the manifest is the route measured. The task maps and analysis plan were frozen before the clean primary lineage; transport-calibration responses are retained separately and never enter the leaderboard.

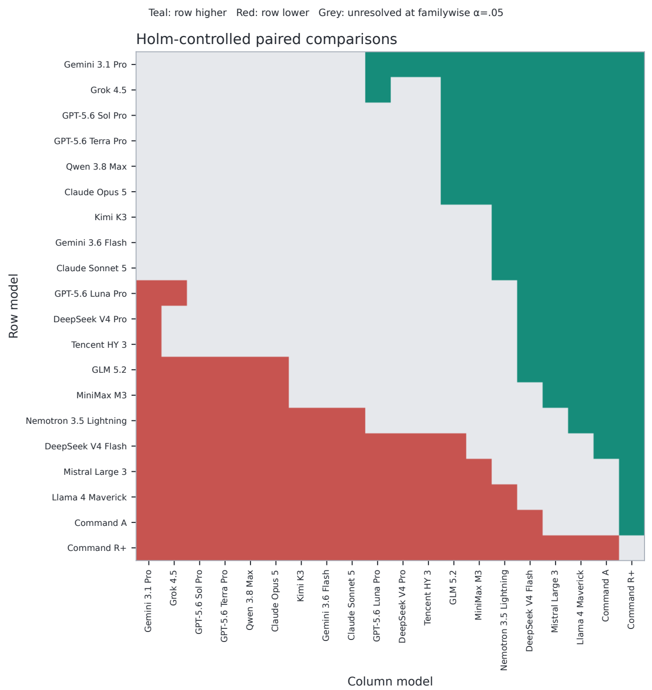

Figure 3: All paired model contrasts after Holm correction. Teal means the row model scores higher; red means lower; grey is unresolved.

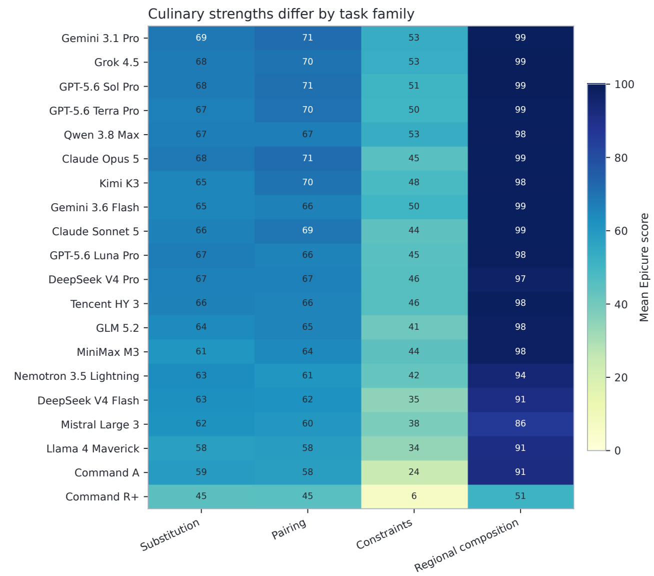

Figure 4: Mean score by task family. Every cell summarizes 160 tasks; values are on the common 0–100 scale but retain family-specific decision semantics.

7 Related work

General-purpose evaluation spans broad suites such as HELM [10], difficult knowledge tests such as MMLU-Pro and GPQA [16, 14], live contamination- resistant sets [17, 7], and human preference systems such as Chatbot Arena [2]. Model judges increase scale but introduce their own bias and calibration problems [20]. FlavourBench instead takes the executable- environment route used by software and agent benchmarks: SWE-bench runs repository tests [8], BFCL evaluates function calls [11], and τ-bench checks interaction outcomes against database state [19].

Culinary datasets and benchmarks cover ingredient networks, recipe understanding, planning, nutrition, and food knowledge [1, 5, 15, 3, 18, 6, 4, 9]. Our focus is narrower and complementary: common-task frontier model measurement against a versioned culinary decision environment, with exhaustive partial-credit maps and paired inference.

8 Limitations

FlavourBench Score measures alignment with one published Epicure release, not universal culinary quality or human preference. Within-task min–max normalization makes the 0–100 scale comparable while discarding absolute raw-utility magnitudes. The tasks evaluate constrained ingredient selection rather than full recipe generation, sensory execution, safety advice, or long-horizon kitchen planning. Endpoint results bind particular provider routes and collection dates; future reruns should be versioned rather than merged into this release. Finally, the dense reward surface can support learning experiments, but the present paper evaluates models and does not demonstrate that optimizing Epicure reward transfers to human cooking outcomes.

9 Conclusion

FlavourBench makes culinary reasoning measurable without asking one language model to grade another. A versioned runtime enumerates every candidate decision, a 20-model common-task panel produces a single continuous score, and paired uncertainty distinguishes resolved differences from apparent rank order. The result is an automated leaderboard that can be rerun, audited, and used as a dense training environment. Its central claim is deliberately concrete: under this executable culinary ground truth and frozen 640-task distribution, the released scores and statistical groups are fully reconstructable from public artifacts.

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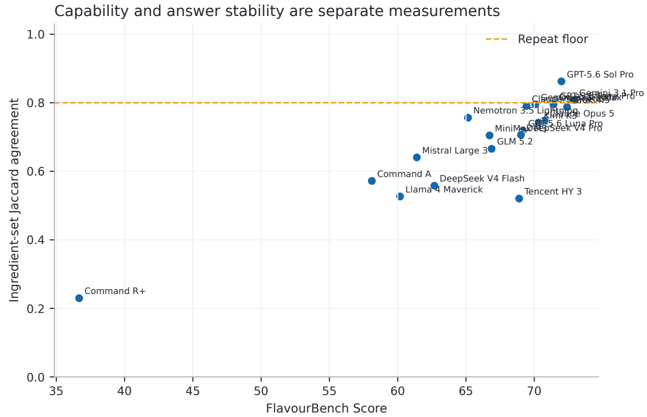

Figure 5: Primary score versus ingredient-set agreement under label permutation. The dashed line is the preregistered 0.80 repeatability requirement for a definitive top-model claim.

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A Exact execution routes

Model	Backend	Frozen route
Gemini 3.1 Pro	openrouter	google-vertex/global/flex
Grok 4.5	openrouter	xai/zdr
GPT-5.6 Sol Pro	openrouter	openai/flex
GPT-5.6 Terra Pro	openrouter	openai/flex
Qwen 3.8 Max	openrouter	alibaba
Claude Opus 5	openrouter	amazon-bedrock/claude-on-aws
Kimi K3	openrouter	morph
Gemini 3.6 Flash	openrouter	google-vertex/us
Claude Sonnet 5	openrouter	amazon-bedrock/claude-on-aws
GPT-5.6 Luna Pro	openrouter	openai/flex
DeepSeek V4 Pro	openrouter	coreweave/fp8
Tencent HY 3	openrouter	atlas-cloud/fp8
GLM 5.2	openrouter	coreweave/fp4
MiniMax M3	openrouter	together
Nemotron 3.5 Lightning	openrouter	coreweave/bf16
DeepSeek V4 Flash	openrouter	deepinfra/fp4
Mistral Large 3	openrouter	mistral
Llama 4 Maverick	openrouter	digitalocean
Command A	openrouter	cohere
Command R+	openrouter	cohere

Table 5: Frozen execution route for every scored model. Automatic fallback is disabled.

B Per-family scores

Model	Substitution	Pairing	Constraints	Regional
Gemini 3.1 Pro	68.5	71.1	53.3	98.7
Grok 4.5	67.7	69.7	53.1	99.0
GPT-5.6 Sol Pro	67.9	70.5	50.7	98.8
GPT-5.6 Terra Pro	66.7	70.3	50.4	98.5
Qwen 3.8 Max	67.0	67.1	53.4	98.2
Claude Opus 5	68.0	71.1	45.3	98.6
Kimi K3	64.9	69.9	48.3	98.1
Gemini 3.6 Flash	65.1	66.3	50.2	98.7
Claude Sonnet 5	66.1	69.1	44.0	98.7
GPT-5.6 Luna Pro	67.4	65.7	45.1	98.4
DeepSeek V4 Pro	66.6	66.7	45.7	97.1
Tencent HY 3	66.1	65.6	45.5	98.5
GLM 5.2	63.9	65.4	40.5	97.7
MiniMax M3	60.6	64.1	44.4	97.8
Nemotron 3.5 Lightning	63.4	60.8	42.5	94.0
DeepSeek V4 Flash	62.8	61.9	35.1	91.0
Mistral Large 3	61.5	60.4	38.0	85.7
Llama 4 Maverick	57.6	58.3	33.8	91.0
Command A	58.9	58.2	24.0	91.3
Command R+	44.7	45.1	6.1	50.9

Table 6: Mean score on each 160-task family. The equal-family mean is the FlavourBench Score.

C Protocol constants

Quantity	Frozen value
Models	20
Primary tasks per model	640
Tasks per family	160
Candidate ingredients	8
Selected ingredients	3
Scored portfolios per task	56
Primary response cells	12,800
Repeat tasks per model	64
Repeat response cells	1,280
Bootstrap replicates	50,000
Sign-flip replicates	100,000
Pairwise hypotheses	190
Ranked completion floor	608/640

Table 7: Core design and inference constants fixed before clean primary execution.